

TIP4PATLIBS

Install an assistant. Then ask better questions.

You do not have to learn SQL. You have to install something that writes it — and then decide what to ask.

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github.com/mtcberlin/epo-tip4patlibs

Nobody handed us a course. We walked into TIP together.

Several walking-group sessions with colleagues from the PATLIB community, working out access to the Technology Intelligence Platform side by side — not a product demo, but people trying to get a first query to return rows.

The same questions, every time

What survives a restart. How to reach PATSTAT. Where a query even starts. How to show a result to a colleague who does not code.

Written down as it worked

Each answer became a notebook that runs top to bottom inside TIP — no installation, and a sensible default so it works before you change anything.

Six modules, one repository

In the order we found them. Everything shown over these two days is in there, public, yours to run and to change.

Before anything else: what these two words mean

1 PATSTAT

the EPO's raw patent database

Not a search interface. Around forty linked tables holding every application, applicant, classification, family link and citation the offices have published.

There is no search box and no company record. You ask it questions in SQL — and that is exactly why it stayed a specialist tool.

2 An AI coding assistant

installed inside your own TIP session

A program that reads your question in plain English, writes the SQL, runs it against PATSTAT, and puts both the code and the result on your screen.

It is not a chatbot bolted onto a website. It sits in your session, in your notebooks, with your data — and you read what it wrote.

Nothing in these two days asks you to write SQL. Everything in them asks you to read a result critically.

The barrier moved.

For twenty years the hard part of patent analytics was the SQL and the schema. That part is now something you can hand over. What is left is the part that was always the professional work.

What you can hand over

- Writing and debugging the SQL
- Knowing which of the forty tables to join
- Remembering the column names and the codes
- The hours between a question and a first table

What stays yours

- Deciding what is actually being asked
- Defining the corpus — and its exclusions
- Judging whether the answer holds
- The caveat you put under the table

Six modules follow. Every one of them is a live notebook in TIP — the slides only set it up.

One claim, three examples, two use cases.

Rising ambition from left to right. Each module is set up on two slides, then shown live in TIP.

THE CLAIM

- 1 Start with TIP**
install an assistant that survives the restart

THREE EXAMPLES

- 2 Query Library**
a question you are actually asked
- 3 PATSTAT Explorer**
the same query, as an app
- 4 Regional Analysis**
your region, your client list

TWO USE CASES

- 5 Antibiotics Landscape**
a multi-step analysis, end to end
- 6 IPScore with TIP**
what is this patent worth?

Modules 1–4 Arne Krüger · mtc.berlin

Modules 5–6 Riccardo Priore · Centro PATLIB, AREA Science Park

1

Install the assistant, persistently

1_startwithtip/

“You have a login to a machine you do not own, which is rebuilt without warning, and which holds the largest patent database in the world. How do you get an assistant onto it that is still there next session?”

WHY THE OBVIOUS ANSWER FAILS

“I will just install it.” The usual install and configuration is gone by Monday — nothing was deleted, the machine was simply rebuilt.

“Then I will learn SQL and PATSTAT.” That was the barrier for twenty years. It is the one part that has stopped being hard.

“The assistant will get it wrong.” Sometimes it will. Which is why the SQL it wrote stays on the screen — and you read it.

LIVE IN TIP NEXT

The persistent setup surviving a restart, then a first PATSTAT query — company and institution search — written by the assistant, read by us.

1 Module 1 — what you have at the end

1_startwithtip/

WHAT NOW EXISTS

- An AI assistant that is still installed after the machine restarts
- One sample query you did not write — and can read, run and change
- A setup you can carry into your own PATLIB's TIP account

THE ONE SENTENCE TO REMEMBER

The barrier was never the database. It was the SQL and the PATSTAT knowledge — and that is the part you can now hand over.

Full 45-minute version of this block: [9_documentation/course/](#)

2

The Query Library

2_querylib/

“A client walks in and asks: who in Europe is working on solid-state batteries?”

WHY THE OBVIOUS ANSWER FAILS

The first answer was always a keyword search. It is not wrong — it is simply the weakest instrument you have.

Keywords miss everything phrased differently, and catch everything that merely mentions the term in passing.

Neither problem is visible in the result. You get a list, and no way to show that it is complete.

LIVE IN TIP NEXT

One query from the library run against PATSTAT Global, then the same query with a single parameter changed.

2

Module 2 — what you have at the end

2_querylib/

WHAT NOW EXISTS

- One query adapted to a question you are actually asked, with its result table
- A first feel for what a PATSTAT query costs in time
- The vocabulary to explain why keywords were combined with classifications

THE ONE SENTENCE TO REMEMBER

An analysis you cannot understand is
one you cannot defend.

Full 45-minute version of this block: [9_documentation/course/](#)

3

PATSTAT Explorer

3_patstat_explorer/

“A client asks how big Siemens Healthineers' patent portfolio is. You run a name search. You get 200 rows. Which one is the answer?”

WHY THE OBVIOUS ANSWER FAILS

None of them. PATSTAT does not know companies — it knows names, exactly as typed on each application.

One entry per spelling, per country, per subsidiary, per typo. There is no company record to look up.

Take the largest row and you have undercounted the group — as a clean, confident-looking number.

**LIVE IN TIP
NEXT**

A name search returning 200 rows, the consolidation decision behind the number, and the same search running as an app inside TIP.

3 Module 3 — what you have at the end

3_patstat_explorer/

WHAT NOW EXISTS

- One consolidated applicant group — a named list of spellings you decided belong together
- The group's family count, filing offices and technology profile
- A written note of what your consolidation missed

THE ONE SENTENCE TO REMEMBER

Consolidation is a judgement, not a lookup. What makes the number defensible is that you wrote the decision down.

Full 45-minute version of this block: [9_documentation/course/](#)

4

Regional Lead Generation

4_lead_generation/

“Your PATLIB is asked to justify its outreach budget. Which companies in your region should you actually be talking to — and why those?”

WHY THE OBVIOUS ANSWER FAILS

Every PATLIB has a contact list. Almost none of them can say how it was built.

Who walked in, who came to the last event, who somebody happened to remember.

The question a ranked list has to survive is not “who is on it” but “who is missing”.

LIVE IN TIP NEXT

The regional notebook on Alsace (FR42), then the same run for a German region — and the tier grid it produces.

4

Module 4 — what you have at the end

4_lead_generation/

WHAT NOW EXISTS

- A named shortlist for your region — family count, depth tier, reach tier, zones
- The grid showing how your region distributes across the tiers
- One defensible sentence about what the list does not contain

THE ONE SENTENCE TO REMEMBER

This is the EP/PCT-active subset of your region. Roughly 70% of national families never take that route — say so under the table.

Full 45-minute version of this block: [9_documentation/course/](#)

5

Patent landscape reports

5_patentreports/

Landscape analyses after Riccardo Priore, Centro PATLIB, AREA Science Park

“A regional cluster manager asks for a picture of what is happening in antibiotic resistance. Where do you start?”

WHY THE OBVIOUS ANSWER FAILS

“Drug resistance” is dominated by cancer — an oncology landscape wearing the wrong label.

Bare acronyms — MRSA, VRE, ESBL — are noisy, and turn up where the subject is not.

Neither is visible in the result: a large corpus, looking perfectly healthy, about something else.

LIVE IN TIP NEXT

The assembled report with its 13 charts — then back to the search strategy that defined the corpus.

5

Module 5 — what you have at the end

5_patentreports/

WHAT NOW EXISTS

- One self-contained HTML report, 13 charts, no internet connection needed
- One workbook carrying the data behind every chart
- A written search strategy — including its exclusions

THE ONE SENTENCE TO REMEMBER

The analyses are generic. The corpus definition is the part that is yours, and the part you have to defend.

Full 45-minute version of this block: [9_documentation/course/](#)

6 What is this patent worth?

6_ipscore_rebuild/

IPScore is an EPO tool; this is the course's own implementation of the 3.01 model

“A client puts a granted European patent on the table and asks what it is worth.”

WHY THE OBVIOUS ANSWER FAILS

Renewal costs, comparable licence rates, “it depends” — or the answer that the question cannot be answered.

IPScore does answer it: 40 questions, seven figures from the accounts, a ten-year discounted cash flow.

And then: all 40 answers come from a person. The engine is exact; its inputs are opinions.

**LIVE IN TIP
NEXT**

The questionnaire and the NPV for one granted European patent, and which of the 40 answers the data could check.

6 Module 6 — what you have at the end

6_ipscore_rebuild/

WHAT NOW EXISTS

- A verified implementation of the EPO IPScore model, seen from the inside
- One completed valuation of a real patent — NPV and profile
- A statement of how much of that number is actually evidence

THE ONE SENTENCE TO REMEMBER

None of the 11 questions PATSTAT can check is one of the 8 that carry money. What a database can check and what a valuation depends on barely overlap.

Full 45-minute version of this block: [9_documentation/course/](#)

The demo is the repository.

1

An assistant on TIP that survives the restart

and writes the SQL you were told you had to learn

2

Three working examples, in rising order of ambition

a ready question · the same query as an app · a named shortlist for your own region

3

Two use cases built the same way

a landscape report, and a patent valuation

4

And in every one of them, the part no assistant does for you

deciding what to ask, and judging whether the answer holds

Seven handouts — the full 45-minute version of every block you just saw — in 9_documentation/course/github.com/mtcberlin/epo-tip4patlibs